

MECH3431 MINI PROJECT

Participants:

Yau Ting Chi 3036222551

Zhao Zehui 3036064052

Zurita Pimentel Adriana Cecilia 3036138499

Introduction

This mini project considers a fictitious office building. A preliminary design for the plumbing and drainage service including cold and flush water supply, soil and waste (S&W) system and stormwater system. Detailed calculation and the conceptual schematic line diagrams and P&D layout plan are all included in the **appendix**. Engineering assumptions are made for calculation.

1. Cold and Flush water supply system

1.1. Sanitary fitments

Considering the usable floor area and the indications stated in the CAP123i, the required number of sanitary fitments for the toilets was calculated. Additionally, a sink is available in the tearoom on each floor.

	Male Toilet	Female Toilet	Accessible Toilet	Tearoom
Basins	3	3	1	
Sinks				1
Water Closets	3	4	1	
Urinals	2			

The toilets and tearoom are located at the core of the building, on the west side, next to the lifts and stairs. This layout allows for a clear distinction between the workplace and the restrooms but keeps the places accessible for the people in the office. Additionally, there are two pipe ducts, one next to the male toilet and another next to the female toilet. This allows for better use of space since the different pipes can be located at the most convenient location to reduce unnecessary length.

1.2. Fresh Water Supply System

The building has 20 floors plus a basement, making its height larger than 77m. Considering this, an indirect supply system would be recommended, as indicated by the WSD. A sump and pump system was selected by taking into account the availability of the two mechanical rooms in the building: B/F and 20/F. The pump at the basement level is able to pump water to the plant room in 20/F, where there is a storage tank. From there, water is distributed to the lower floors through a gravity-fed system for floors 1/F to 16/F. For floors 17/F to 20/F, a booster system was adopted. This is because the water tank is located at the same level as the floors it serves, meaning that the pressure in the pipe cannot reach the minimum required for the water points by itself. A single jet water meter

is installed at the corresponding position for billing and monitoring purposes as dictated by the WSD.

The design and sizes of the pumps, tanks, and pipes used here correspond to those indicated by the WSD and IOP. The roof tank has a capacity of 5400 L and is responsible for the overall demand of the building. If the tank reaches a low level, the pump in the basement will pump from the sump tank (1800 L) to refill it and ensure water is provided without interruptions. For the pneumatic booster at 20/F, it was assumed that the minimum pressure at the waterpoint is 10mH. Additionally, it is needed to install a Pressure Reducing Valve between 4/F and 3/F to avoid excessive built-up pressure because as the water descends through the first 16 floors, it approaches the assumed maximum pressure at a waterpoint of 50mH. For further details on the calculations of pipe size and pressure at each floor and fitment, please refer to the Appendix and Schematic Diagrams.

1.3.Flush Water Supply System

The design of the flush water supply system is similar to the freshwater supply system. The flush water from the mains is first stored in the 2025 L tank in the basement. When needed, the pump transfers the water to the roof tank of 6075 L. From there, flush water is distributed by gravity or by the booster pump system. The pressure requirement is similar to that of freshwater. Since the flush water uses PVC pipes instead of copper, the pipe sizing is different. A PRV is also required in this system. Additionally, a turbine meter is included for monitoring purposes, as suggested by the WSD.

1.4. Pump operation

The pump at B/F will operate when the level of the roof tank is detected to be low by the sensor. The pump at 20/F will operate as required by the consumption on the upper four floors. Both pumps were designed following regulations, and the full calculations are displayed in the Appendix.

1.5. Further Considerations

- In order to make this building eco-friendly, water taps with infrared sensors are used, following the recommendation by WSD. Similarly, sanitary fitments with a Grade 1 on the WELS will be preferred for installation.
- It is advised to regularly check both meters to monitor consumption and identify the proper efficiency of the system. Regular inspections of the pumps and pipes are recommended to ensure the system is working properly.
- Due to the complex setup of the pipes, a false ceiling will be used to allow for the pipes to pass through the different rooms while keeping the aesthetics of the building.
- Since the upper mechanical room is located at the same level as the offices, soundproofing of the plant room will be required to prevent any inconvenience for the people in the building. Additionally, it must be ensured that the mechanical room is easily accessible for maintenance, which requires the cooperation of the people in the office to keep a tidy space.

2. S&W drainage system

For the solid and waste drainage system, the above-ground drainage system and the below-ground drainage system are designed separately from the rainwater system. The detailed calculation and schematic diagrams were shown in the **appendix**. A cast iron pipe will be used for the underground drainage system.

2.1. Above-ground solid and waste system

Concerning the aboveground S&W system, toilets are considered from 1/F to 20/F. The main stack is designed to be $\phi 150\text{mm}$ made of PVC, together with the $\phi 50\text{mm}$ vent stack for the first 10 floors, and a $\phi 100\text{mm}$ made of PVC main stack, together with the $\phi 50\text{mm}$ vent stack for the next 10 floors. The termination point of the pipe for both the main stack and vent stack will be designed at above 1.5m of the R/F to prevent any foul gases transmitted to the green roof. Cross vents will be installed on every 3 floors to minimize pressure fluctuation to improve the water seal loss prevention. The soil stack and the vent stack will be installed in the pipe duct. As a result, the installation cost of the pipe can be minimized.

2.2. Below-ground solid and waste system

Concerning the belowground S&W system, the pipe is designed with an inclination of 1:30 made of cast iron. A manhole is installed where the pipe turns from the aboveground to the underground. A petrol interceptor with 3 compartments is installed for removing oil and grease from wastewater and provides a time of retention for the oil and water to separate before the water is discharged.

3. Stormwater drainage system

According to the Building Ordinance (CAP123I regulation), every 700 mm^2 of pipe cross-section area shall be provided for every 10 m^2 of horizontal roof area; the diameter of stormwater downpipe shall be 600mm minimum. No venting or water sealing is needed. The return period of the rainwater is assumed to be 20 years. Rainwater will be collected by the roof and the carpark, the material of the both location is concrete. Stormwater pipes will be installed underground, along the roof and at the outside of the building for evenly distribution, detail layout is shown in **appendix**. This design includes 14 stormwater downpipes with a diameter of 80mm. Material for aboveground stormwater pipe will be PVC while for underground pipes will be cast iron.

The stormwater downpipes will be merged into 2 underground stormwater pipes with a diameter of 200mm and an inclination of 1 in 70 to ensure smooth water flow. There will be rainwater collected from the carpark. Therefore, rainwater pit and submersible pump will be installed to pump the rainwater to the last manhole above the basement. Submersible pump on/off control is controlled by the level switch inside the rainwater pit.

Underground stormwater pipes will be installed under G/F and above B/F. The underground stormwater pipes are directly connected to the last manhole located at the southeast corner of the building.

4. Conclusion

In this project, a preliminary design for the plumbing and drainage services of a factitious office building is performed, including cold and flush water supply system, soil and waste (S&W) system and stormwater system. The detail for this drainage system, including the sizing of pipework, tanks and equipment, and system demand is considered.

5. References

1. Plumbing Engineering Services Design Guide, The Institute of Plumbing
2. Technical Requirements for Plumbing Works in Buildings, Water Supplies Department (WSD)
3. Laws of Hong Kong, CAP123I, HKSAR Government

Appendix

Calculation for Fresh Water Supply

1. Determining sanitary fitments according to Building Regulations (CAP123i)

- Usable floor area = 917.6 m^2 . Population density = 1 person/ 9 m^2 . Hence, population per floor = 102 people.
- Ratio of male to female = 1:1. Hence, male to female = 51 male:51 female.
- Sanitary fitments per floor:

	Male Toilet	Female Toilet	Accessible Toilet	Tearoom
Wash Basin	3	3	1	
Sink				1

2. Determining the total water storage capacity and allocation of water tanks

- Total number of water points: $(3+3+1+1) \times 20 = 160$ points. Design practice of 45L/point. Total cold-water storage = $160 \times 45 = 7200 \text{ L}$.
- Roof tank (20/F): sump tank ratio = 3:1 according to the Plumbing Installation Guidelines by WSD. For a total of 7200 L, Roof tank: Sump tank = 5400 L:1800 L

3. Determining the Simultaneous Demand of the various parts of the system

- Overall Simultaneous Demand. N° of basins and sinks = $160 \times 2 \text{ LU}$ (medium frequent use) = 320 LU. Overall Simultaneous demand = 2.7 L/s (from the copper pipe graph)
- Simultaneous Demand for Gravity Feed, excluding top 4 floors. 8 water points/floor $\times$ 16 floors = 128 basins and sinks. $128 \times 2 \text{ LU} = 256 \text{ LU} = \underline{2.3 \text{ L/s}}$.
- Simultaneous Demand for Top Floor Booster System. 8 water points/floor $\times$ 4 floors = 32 basins and sinks. $32 \times 2 = 64 \text{ LU} = \underline{0.9 \text{ L/s}}$
- Simultaneous Demand for Tee-off. For each male and female toilet, assuming vertical pipe tees off separately: 3 basins $\times$ 2 LU = 6LU = 0.23 L/s. For each accessible toilet: 1 basin = 2 LU = 0.13 L/s. For the tearoom: 1 sink = 2LU = 0.13 L/s

4. Sizing the Up-feed Pump and Pipe

- Up feed pump flow rate must serve the overall simultaneous demand. Up feed pump flow rate = 2.7L/s
- Pipe sizing. By following the graph for copper pipe, we should choose the 42mm pipe, resulting in a head loss of 0.13mH per m run. The velocity is approximately 2.2 m/s, which is less than 3 m/s, hence, it is suitable.
- Pump head calculation. Pump head = Static head + Head loss due to friction and fitting. Static Height = 20 floors $\times$ 3.4 m + 1 G/F $\times$ 5m + 1 B/F $\times$ 4.5 m = 77.5m. Assume pipe length = 90m by considering some of the pipe runs horizontally. Equivalent length = $90 \times 1.3 = 117 \text{ m}$, assuming 30% for fitting losses. Total head loss = $117 \times 0.13 = 15 \text{ mH}$. Pump head = $117 + 15 = 132 \text{ mH}$.

5. Sizing the down feed supply pipes

- a. For the gravity-fed system. Assume head loss of 0.1mH per m run using a copper pipe. Based on Graph 3, from the IOP Plumbing Engineering Services Design Guide, the individual carrying capacities of the Cu pipe are as follows:

Diameter (mm)	Flow rate (L/s)	Loading Unit (LU)
54	8.8	1450
42	4.9	600
35	2.4	220
28	1.4	120
22	0.8	50
15	0.13	2

The carrying capacities of the individual pipes are calculated.

From	To	Loading Units	Pipe Size (mmØ)
20/F (first tee off at 17/F)	14/F	256 to 224	42
14/F	13/F	224 to 208	42 to 35
13/F	8/F	208 to 128	35
8/F	7/F	128 to 112	35 to 28
7/F	4/F	112 to 64	28
4/F	3/F	64 to 48	28 to 22
3/F	1/F	48 to 16	22

- b. For the top-floor booster system. Following the same approach as above.

From	To	Loading Units	Pipe Size (mmØ)
20/F (booster)	20/F	64	35
20/F	17/F	48 to 16	28

- c. Sizing of the main tee off pipe.

	Simultaneous Demand L/s	Pipe Size (Ømm)
Male Toilet	0.23	22
Female Toilet	0.23	22
Individual Basin	0.13	15
Individual Sink (tearoom)	0.13	15

6. Sizing the Top-floor Booster System

- a. Pump sizing. Simultaneous demand = 0.9 L/s. Assume a minimum head requirement at the water point = 10mH. Head gain = 0m as the booster is at the 20/F, which is also the floor it is serving. Assume a horizontal length of 15m. Accounting for fitting losses, overall hydraulic distance = 15×1.3

= 20. Since the pipes are based on 0.1mH/m run, overall head loss = $20 \times 0.1 = 2.0\text{mH}$. Pump head + static head gain - fitting and friction loss = residual pressure. Pump head + 0 - 2 = 10. Hence, the Pump head (Cut-in pressure) = 12mH at the flow rate of 0.9 L/s.

- b. Pneumatic Tank Sizing. Assume the difference between cut-in and cut-out = 15mH. Hence, cut-out pressure = 27 mH. Assume cut-out time = 5 min. Effective volume of the tank = $0.9 \times 5 \times 60 = 270\text{ L}$. By using Ideal Gas law: $P_2 \times V_2 = P_3 V_3$, where $P_2 = 215700\text{ Pa}$ (12mH gauge at pump cut-in + 10mH atm. Pressure), $P_3 = 362800\text{ Pa}$, and $V_2 - V_3 = 0.27\text{ m}^3$. $215700 \times (0.27 + V_3) = 362800 \times V_3$. Hence, $V_3 = 0.40\text{ m}^3$. Assume the tank is not pre-charged. $P_1 \times V_1 = P_3 \times V_3$. $98000 \times V_1 = 362800 \times 0.40$. Hence, $V_1 = 1.5\text{ m}^3$.
Volume of pneumatic water tank = 1500L

7. Location of Pressure Reduction Valve

- a. For the gravity-fed system. Built up static pressure per floor = 3.4mH. Vertical pipe length = 3.4m. Loss at vertical pipe = $0.1 \times 3.4 = 0.34\text{mH}$. Hence, during operation, built-up pressure per floor = $3.4 - 0.34 = 3.06\text{ mH}$. Assume the maximum pressure allowed at a water point is 50mH. The approximate pressure of water at 20/F is 0mH. Number of floors required to create a pressure built-up of 50mH = $50/3.06 = 16.3$. Take 16 floors. Hence, PRV should be installed between 4/F and 3/F. Anticipated pressure at PRV = 49 mH. Consider using a PRV with a pressure ratio of 3:1, pressure at the downstream of the PRV = 16.3 mH. No further PRVs are needed as pressure does not build up past 50mH again.

8. Connection to the Government Mains

- a. Sizing the pipe. The pipe connecting the building's water supply system to the Government mains must be able to serve the overall simultaneous demand of the whole building. Similar to the calculation in 4. b), we should choose the 42mm copper pipe. Considering that the water mains pipe is 100mm in diameter, a pipe adapter should be used to make the connection between the mains and the building services. The pressure provided by the government mains is 20mH, which is higher than the minimum residual pressure required of 10mH. Hence, this pressure is acceptable.
- b. Metering. Based on the size of the pipe, we can determine that the most appropriate type of water meter is a single jet water meter, according to the WSD. Since the connection is bigger than 40mm, it is also required to install a gate valve before the meter position and a check valve after the meter.

Calculation for Flush water supply

1. Determining sanitary fitments according to Building Regulations (CAP123i)

- a. Based on the Fresh Water Supply section, we know the ratio male:
female = 51:51.
- b. Sanitary fitments per floor

	Male Toilet	Female Toilet	Accessible Toilet
WC	3	4	1
Urinal	2		

2. Determining the total water storage capacity and allocation of water tanks

- a. Total number of water points = $(3+4+1+1) \times 20 = 180$ points. Design practice of 45L/point. Total Flush water storage = $180 \times 45 = 8100\text{L}$
- b. Roof tank (20/F): sump tank ratio = 3:1 according to the Plumbing Installation Guidelines by WSD. For a total of 8100L, Roof tank: Sump tank = 6075L:2025L

3. Determining the Simultaneous Demand for various parts of the system

- a. Overall Simultaneous Demand. N° of WCs = $8 \times 20 \times 2\text{LU} = 320\text{LU}$. N° of urinals = $2 \times 20 \times 1\text{LU} = 40\text{LU}$. Total loading units = 360LU. Overall Simultaneous Demand = 3.2 L/s (from the PVC pipe graph).
- b. Simultaneous Demand for Gravity Feed, excluding top 4 floors. N° of WCs = $8 \times 16 \times 2\text{LU} = 256\text{LU}$. N° of urinals = $2 \times 16 \times 1\text{LU} = 32\text{LU}$. Total LU for Gravity Feed = 288LU. Total demand = 2.6 L/s
- c. Simultaneous Demand for Top Floor Booster System. N° of WCs = $8 \times 4 \times 2\text{LU} = 64\text{LU}$. N° of urinals = $2 \times 4 \times 1\text{LU} = 8\text{LU}$. Total LU = 72LU. Total demand = 1.0 L/s
- d. Simultaneous Demand for Tee-off. At the male toilet, 3 WCs $\times 2\text{LU} + 2$ urinals $\times 1\text{LU} = 8\text{LU} = \underline{0.27 \text{ L/s}}$. At the female toilet, 4 WCs $\times 2\text{LU} = 8\text{LU} = \underline{0.27 \text{ L/s}}$. At the accessible toilet, 1 WC $\times 2\text{LU} = 2\text{LU} = \underline{0.12 \text{ L/s}}$

4. Sizing the Up-feed Pump and Pipe

- a. The Overall Simultaneous Demand of the building must be served by the up-feed pump when the storage is consumed. Hence, the up-feed pump flow rate = 3.2 L/s
- b. Pipe sizing. According to the graph for PVC pipe, we should choose the 63mm pipe, resulting in a head loss of 0.06mH per m run. The velocity is approximately 1.7 m/s, which is less than 3 m/s; hence, it is suitable.
- c. Pump head calculation. Pump head = Static head + Head loss due to friction and fittings. Static Height = 20 floors $\times 3.4 \text{ m} + 1 \text{ G/F} \times 5 \text{ m} + 1 \text{ B/F} \times 4.5 \text{ m} = 77.5 \text{ m}$. Assume pipe length = 90m by considering some of the pipe runs horizontally. Equivalent length = $90 \times 1.3 = 117 \text{ m}$, assuming 30% for fitting losses. Total head loss = $117 \times 0.06 = 7.02 \text{ mH}$. Pump head = $117 + 7.02 = 124 \text{ mH}$

5. Sizing the Down feed supply pipes

- a. For the gravity-fed system. Assume a head loss of 0.1mH per m run using a PVC pipe. Based on the IOP Plumbing Engineering Services Design Guide:

Diameter (mm)	Flow rate (L/s)	Loading Unit (LU)
63	4.3	520
50	2.4	230
40	1.3	90
32	0.7	39
25	0.35	14
20	0.18	4

The carrying capacities of the individual pipes are calculated.

From	To	Loading units	Pipe size (mmø)
20/F (first tee off at 17/F)	13/F	288 to 234	63
13/F	12/F	234 to 216	63 to 50
12/F	6/F	216 to 108	50
6/F	5/F	108 to 90	50 to 40
5/F	3/F	90 to 54	40
3/F	2/F	54 to 36	40 to 32
2/F	1/F	36 to 18	32

- b. For the top-floor booster system. Following the same approach as before.

From	To	Loading Units	Pipe size (mmø)
20/F (booster)	20/F	72	40
20/F	19/F	54	40
19/F	17/F	36 to 18	32

- c. Sizing of the main tee off pipe.

	Simultaneous Demand (L/s)	Pipe size (mmø)
Male Toilet	0.27	25
Female Toilet	0.27	25
Individual WC	0.12	20
Individual Urinal	0.09	16

6. Sizing of the Top-Floor Booster system.

- a. Pump sizing. Simultaneous demand = 1.0 L/s. Assume a minimum head requirement at the water point = 10mH. Head gain = 0m as the booster is at the 20/F, which is also the floor it is serving. Assume a horizontal length of 15m. Accounting for fitting losses, overall hydraulic distance = $15 \times 1.3 = 20$. Since the pipes are based on 0.1mH/m run, overall head loss = $20 \times$

$0.1 = 2.0\text{mH}$. Pump head + static head gain - fitting and friction loss = residual pressure. Pump head + $0 - 2 = 10$. Hence, the Pump head (Cut-in pressure) = 12mH at the flow rate of 1.0L/s .

- b. Pneumatic Tank Sizing. Assume the difference between cut-in and cut-out = 15mH . Hence, cut-out pressure = 27 mH . Assume cut-out time = 5 min . Effective volume of the tank = $0.9 \times 5 \times 60 = 270\text{ L}$. By using Ideal Gas law: $P_2 \times V_2 = P_3 \times V_3$, where $P_2 = 215700\text{ Pa}$ (12mH gauge at pump cut-in + 10mH atm. Pressure), $P_3 = 362800\text{ Pa}$, and $V_2 - V_3 = 0.27\text{ m}^3$. $215700 \times (0.27 + V_3) = 362800 \times V_3$. Hence, $V_3 = 0.40\text{ m}^3$. Assume the tank is not pre-charged. $P_1 \times V_1 = P_3 \times V_3$. $98000 \times V_1 = 362800 \times 0.40$. Hence, $V_1 = 1.5\text{m}^3$.
Volume of pneumatic water tank = 1500L

7. Location of Pressure Reduction Valve

- a. For the gravity-fed system. Pressure build-up per floor = 3.4mH . Vertical pipe length = 3.4m . Loss at vertical pipe = 0.34mH . Hence, during operation, built-up pressure per floor = $3.4 - 0.34 = 3.06\text{ mH}$. Assume the maximum pressure allowed at a water point is 50mH . The approximate pressure of water at 20°F is 0mH . Number of floors required to create a pressure built-up of $50\text{mH} = 50/3.06 = 16.3$. Take 16 floors. Hence, PRV should be installed between 4°F and 3°F . Anticipated pressure at PRV = 49 mH . Consider using a PRV with a pressure ratio of $3:1$, pressure at the downstream of the PRV = 16.3 mH . No further PRVs are needed as pressure does not build up past 50mH again.

8. Connection to the Government Mains

- a. Sizing the pipe. The pipe connecting the building's water supply system to the Government mains must be able to serve the overall simultaneous demand of the whole building. Similar to the calculation in 4. b), we should choose the 63mm pipe. Given that the government mains are 80mm in diameter, a pipe adapter must be used between the mains and the inside services. The pressure provided by the government mains is 15mH , which is higher than the minimum residual pressure required of 10mH . Hence, this pressure is acceptable.
- b. Metering. Although seawater is not metered, the WSD indicates that a meter position should be included to monitor consumption. Based on the size of the pipes, the best option would be a turbine meter. Additionally, a stop valve should be placed on the inlet side of the meter, and a check valve should be placed after the meter.

Calculation for Soil and Waste Drainage System

1.Calculation for aboveground S&W drainage

Usable floor area = 917.6 m²

No. of floors with toilets: 1-20/F

Population density = 1 person/ 9m²

Thus, population per floor = 102 people

Ratio of male to female = 1:1

Thus, male to female = 51 male:51 female

Schedule of sanitary fitments per floor:

	Male Toilet	Female Toilet	Accessible Toilet	Pantry
Wash Basin	3	3	1	0
Water Closet	3	4	1	0
Urinal	2	0	0	0
Floor drain	1	1	1	1
Sink	0	0	0	1

According to the IOP DU calculation

Each floor		Discharge Units (DU)
Wash basins:	(3+3+1) x 0.3	2.1
Water Closets	(3+4+1) x 1.8	14.4
Urinals	1 x 0.4	0.4
Kitchen sink	1 x 1.3	1.3
		$\sum DU = 18.2$

There are 20 floors containing sanitary fitments; therefore, the total DU is 20 x 18.2 = 400.4.

Concerning usage of the appliance, the K factor is 0.5 (office use).

Therefore, the wastewater discharge $Q_{ww} = K\sqrt{DU} = 0.5 \times \sqrt{400.4} = 10 \text{ L/s}$

Concerning the air conditioning unit with a peak discharge with 0.2 L/s and no pump flow rate discharge in this building.

$$Q_{tot} = 10 + 0.2 = 10.2 \text{ L/s}$$

According to the table from the “Plumbing Engineering Services Design Guide”, a Ø150mm S&W stack with Ø50mm vent stack is selected.

2.Calculation for underground S&W drainage

For a floor of 102 people, the design flow rate is $\frac{1}{2} \times \frac{102 \times 225}{6} = 1912.5 \text{ L/hr} = 0.531 \text{ L/s}$.
The total number of floors is 20, so the flow rate is $0.531 \times 20 = 10.62 \text{ L/s}$.

Assume $\frac{1}{2}$ bore flow for the drainpipe and running at about 1.2m/s, $\frac{1}{2}$ bore flow area = $(10.62/1000)m^3/s \div 1.2 \text{ m/s} = 0.0089m^2$.

$$\text{The pipe diameter is } \frac{1}{2} \times \left(\frac{\pi D^2}{4} \right) = 0.0089,$$

$$D = 0.151m.$$

A commercially available pipe size would be 150mm. Carrying capacity of 150mm pipe at $\frac{1}{2}$ bore and 1.2m/s = $\frac{1}{2} \times \left(\pi \times \frac{0.15^2}{4} \right) \times 1.2 = 0.0106m^3/s$.

The hydraulic mean depth for half full bore = $\frac{0.15}{4} = 0.0375m$.

Using Manning formula, assume $n = 0.017$ for poor cast iron pipe,

The velocity of the flow is $\frac{1}{0.017} 0.0375^{\frac{2}{3}} (i)^{\frac{1}{2}} = 1.2m/s$, therefore, the $i = 0.033$ or 1 in 30

At $\frac{1}{2}$ bore, flow = $0.0106m^3/s$, velocity 1.2m/s for this 150mm pipe inclined at 1:30.

According to the proportional chart by WSD,

For a $\frac{1}{2}$ bore, the proportional depth = 0.5:

The proportional discharge $Q/Q_o = 0.5$, $Q_o = 0.0212m^3/s$.

The proportional velocity $V/V_o = 1.0$, $V_o = 1.2m/s$.

At design discharge = $0.0156m^3/s$:

$$Q/Q_o = \left(\frac{0.0156}{0.0212} \right) = 0.736, \frac{d}{D} = 0.65$$

$$\frac{d}{D} = 0.65, V/V_o = 1.074.$$

The flow velocity is 1.29m/s. Therefore, both the bore and the flow velocity are acceptable.

The minimum flow is 1/6 of the design flow, which is $\frac{0.0156}{6} = 0.0026m^3/s$.

With the same method to check for the Q/Q_o and V/V_o :

$$Q/Q_o = \left(\frac{0.0026}{0.0212} \right) = 0.123; V/V_o = 0.73,$$

Therefore, the velocity is $0.73 \times 1.2 = 0.88\text{m/s}$. Since 0.88m/s is larger than the self-cleansing velocity (0.8m/s), a cast iron pipe in $\phi 150\text{mm}$ with inclination of 1 in 30 is acceptable.

Sizes of the branch pipes and discharge pipes:

	WC	Urinals	Basins
Male Toilet	100mm	50mm	50mm
Female Toilet	100mm	Not Required	50mm

Sizes of the drainpipes for individual sanitary fitments:

WC	Urinals	Basins
100mm	400mm	32mm

Sizes of Vent pipes for individual sanitary fitments:

WC	Urinals
50mm	32mm

Calculation for stormwater drainage

Given information:

1. Area of building = 931m² (based on AutoCAD)
2. Perimeter of building = 136.7m (based on AutoCAD)

According to Buildings Ordinance (Cap. 123I Regulation 32):

- Every 700 mm² of pipe cross-section area shall be provided for every 10 m² of horizontal roof area
- Diameter of stormwater pipe shall be 65 mm minimum

Assumptions:

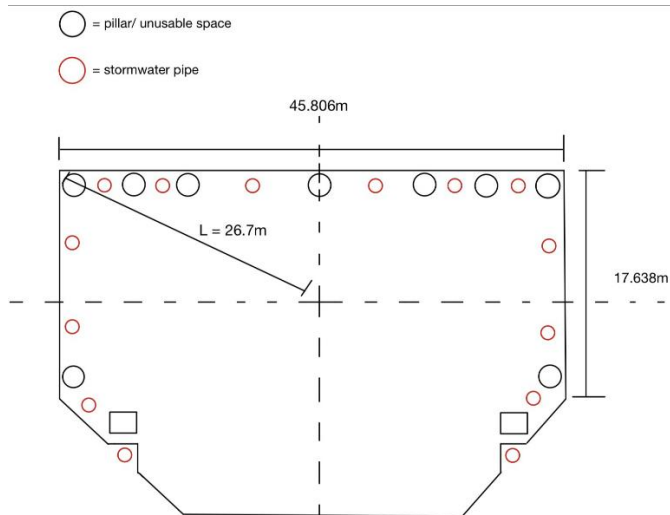
- Each downpipe is sized at 80mm²
- Only the roof is used to collect rainwater, no vertical wall catchment area

$$\text{Downpipe cross-sectional area} = \frac{\pi D^2}{4} = \frac{\pi (80)^2}{4} = 5026.55 \text{ mm}^2$$

Minimum number of downpipes needed to satisfy the Buildings Ordinance

$$\frac{(\frac{931}{10})(700)}{5026.55} = 12.9 \approx 13$$

In order to place the stormwater downpipes evenly, pipes are spaced evenly between the cylindrical pillars. Two of the downpipes are located at the outside of the south side of the building due to the location of the fixed rooms. To make the distribution neater, 14 downpipes are placed instead of 13. The roof layout is as shown below:



By Brandsby William's Equation,

$$t_0 = 0.14465 \times \left(\frac{L}{H^{0.2} A^{0.1}} \right)$$

Let slope be 1:100, $H = 1$

$$A = \frac{45.806}{2} \times \frac{17.638}{2} = 22.9 \times 13.75 = 314.9 \text{ m}^2$$

$$L = \sqrt{22.9^2 + 13.75^2} = 26.7 \text{ m}$$

$$t_0 = 0.14465 \times \left(\frac{(26.7)}{(1)^{0.2}(314.9)^{0.1}} \right) = 2.17 \text{ min}$$

Using IDF relationship,

$$i = \frac{a}{(t_d + b)^c}$$

Assuming that the return period = 20-year,

$$i = \frac{(639)}{(2.17 + (4.3))^{(0.43)}} = 286 \text{ mm/hr}$$

By Rational Method,

$$Q = C \times I \times A$$

Assuming that the roof is concrete, $C \approx 0.9$

$$Q = (0.9) \times \left(\frac{286}{\frac{1000}{3600}} \right) \times (917) = 0.0655 \text{ m}^3/\text{s} \approx 65.5 \text{ L/s}$$

Amount of rainfall each downpipe collects = $65.5/14 = 4.68 \text{ L/s}$

Using vertical stack empirical formula,

$$Q = kD^{\frac{8}{3}}$$

$$(4.68) = k(80)^{\frac{8}{3}}$$

$$k = 3.94 \times 10^{-5}$$

The value of k lies between $1/4$ and $1/3$ flow, thus acceptable.

Given that the invert level is at -1.5m for rainwater drains

Assumptions:

1. Pipe diameter = 200mm
2. Cast iron as the pipe material, thus Manning Coefficient $n = 0.017$
3. Maximum capacity of the pipe is considered, proportional depth = 0.75

By Manning Formula,

$$v = \frac{1}{n} R_h^{\frac{2}{3}} i^{\frac{1}{2}}$$

$$R_h = \frac{D}{4} = \frac{(0.2)}{4} = 0.05\text{m}$$

$$i = \frac{0 - (-1.5)}{(40)} = 0.0375$$

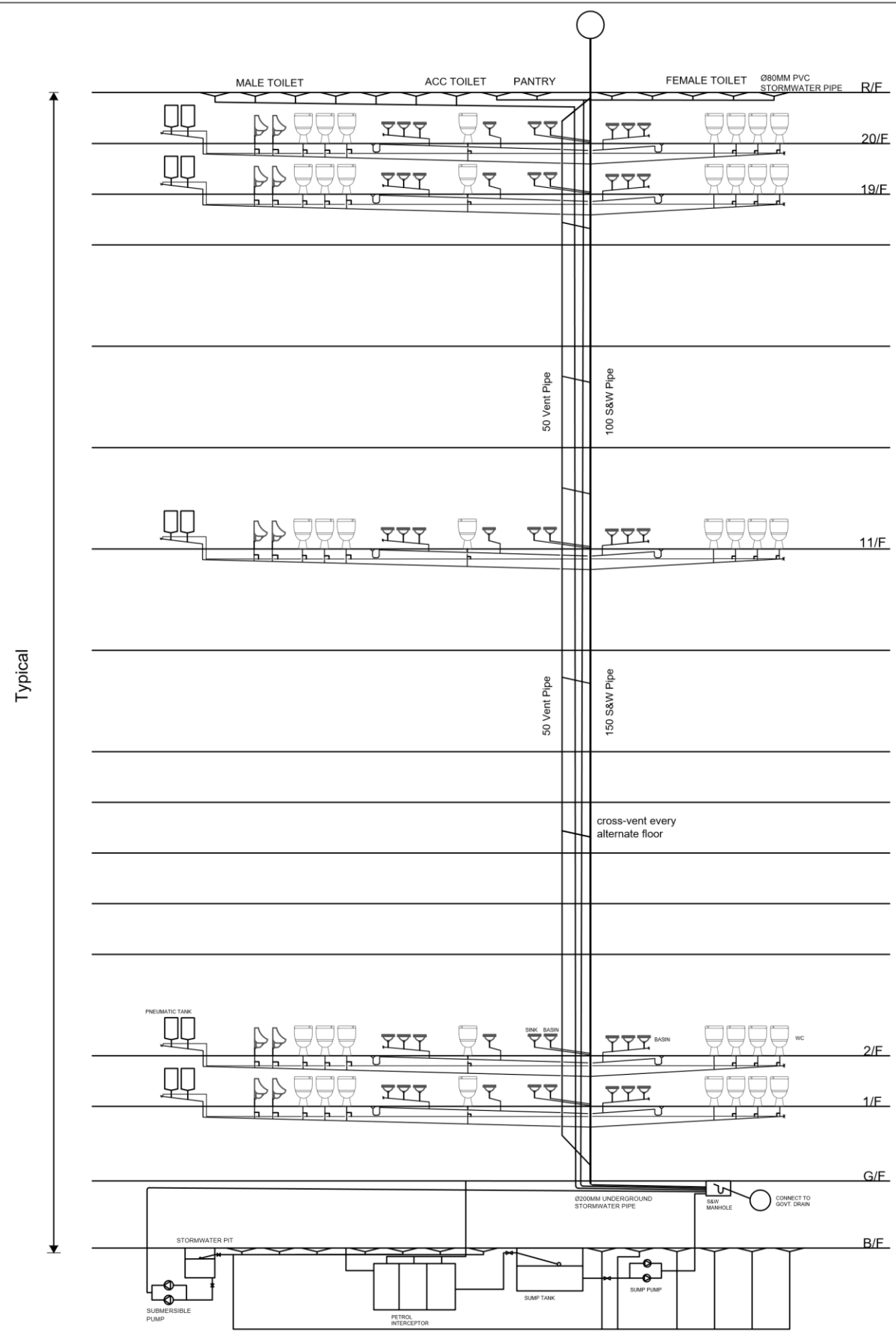
$$v = \frac{1}{(0.017)} (0.05)^{\frac{2}{3}} (0.0375)^{\frac{1}{2}} = 1.546\text{m/s}$$

$$A = \frac{\pi(0.2)^2}{4} \times (0.75) = 0.2356\text{m}^2$$

$$Q = Av = 0.03649\text{m}^3/\text{s} \approx 36.5\text{L/s}$$

$$\text{Peak flow rate} = 65.5\text{L/s}$$

Therefore, at least 2 horizontal 200mm² pipes with an inclination of 1 in 70 are needed for this the rainwater pipes from the roof.



Typical

LEGEND LIST	
	URINAL
	WATER CLOSET
	SINK/ BASIN
	VALVE
	PUMP
	PNEUMATIC TANK

PROJECT

DRAWING TITLE

Schematic Diagram for Rainwater and S&W Disposal System

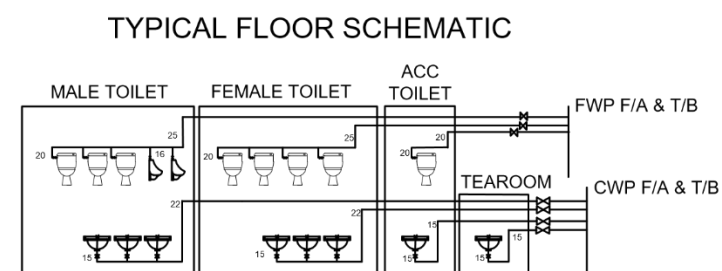
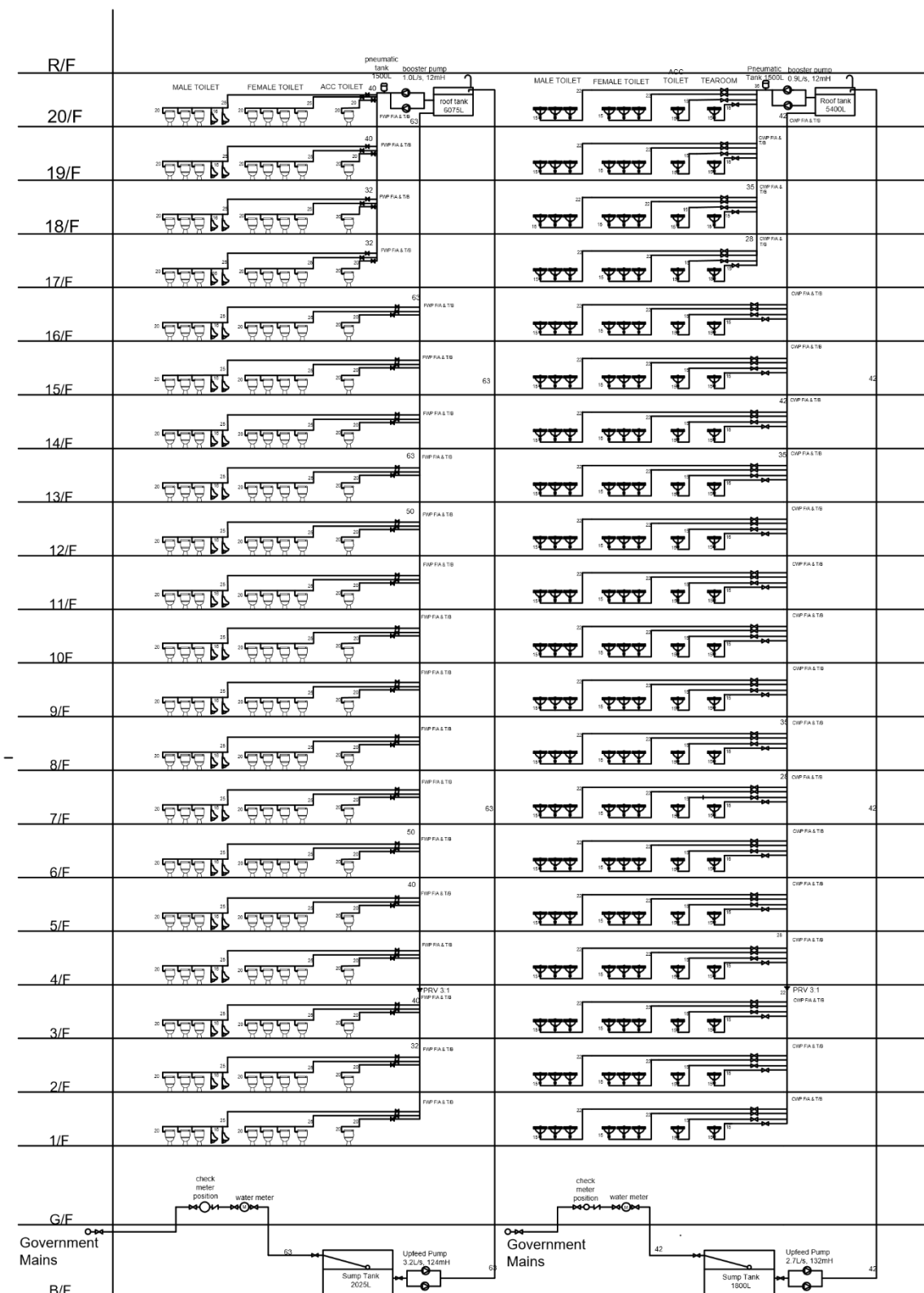
JOB NO

DESIGNED BY DRAWN BY CHECKED BY

APPROVED FOR ISSUE DATE

CAD FILE SCALE

DWG NO



LEGEND LIST	
	URINAL
	WATER CLOSET
	SINK/BASIN
	VALVE
	PUMP

PIPE MATERIALS	
COLD WATER	COPPER
FLUSH WATER	PVC

PROJECT
MINI PROJECT MECH3431
DRAWING TITLE
SCHEMATIC DIAGRAM OF COLD AND FLUSH WATER SYSTEM
JOB NO.
DESIGNED BY DRAWN BY
APPROVED FOR ISSUE
CAD FILE
DWG NO.

